PROJECT PROPOSAL

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Project Title: *The Water Tracker Intake and Questionnaire (With Guaranteed Results!)*

Problem Statement:

Many people forget to drink water sometimes, and often do not even know if they drank enough or exceeded the amount of water to drink for a day. One problem is how much they should drink actually, the classic "drink 8 glasses of water" everyday won't work for everyone. Different people with different lifestyles would need different amount of water intake.

This is where our "The Water Intake Questionnaire" comes in. It calculates how much water the user has to drink depending on factors like weight, exercise frequency, sex, and age. This helps users know exactly how much water they need and if they reached their goal.

The 2005 report on Dietary Reference Intake for Water conducted by the National Academies of Sciences, Engineering, and Medicine concluded that an individual's water requirements differ a lot based on their weight, exercise frequency, sex, and age.

Planned Objectives:

The main goal of our project is to build a functional Python program that helps users keep track of their daily hydration habits. First, we aim to design a system that takes in a user's basic info, like their weight, age, sex, and how often they work out, and uses those specific metrics to calculate a personalized daily water goal. From there, we want to create a continuous loop where users can easily log how many cups they drink and what time they drank them throughout the day. At the end of the day, the program will calculate the final total and compare it to their goal, giving them a clear breakdown of whether they stayed hydrated or fell short.

Yes, there are many other apps designed to help people with hydration. They vary in ways such as how they implement features, how they motivate people to drink, and how they personalize the tracker from person to person. For example, Plant Nanny has a vibrant and colorful theme, and it uses plants to motivate people. Every time they drink, the plant grow, but it withers when they forget to hydrate themselves. To make our own app different or unique from the others, we are going to let the user customize their color theme and overall vibe. We are also going to make our app simple, unlike others which are complicated and hard to navigate.

Planned Features:

- *Enter user's name, their weight, exercise frequency, sex, water goal intake, and age.*

- *Show options for exercise frequency: A. Never B. Rarely C. Normally D. Frequently E. Always*

- *Calculate the base water goal in cups of the user by dividing their body weight in pounds by 16.*

- *Show options for biological sex: M. Male: Adds 3 extra cups to the goal for average muscle mass needs. F. Female: Uses the regular baseline amount.*

- *Shows rules based on age groups: Under 14: Sets small goal because they have smaller bodies. 14-59: Uses the standard baseline based on body weight. 60 and above: Sets a safe minimum goal to protect older adults from dehydration.*

- *Ask the user's water intake and the time of the drink inside a loop until they choose to end the day, to see if they reached their target or if they need to drink more.*

- *Use a variable to hold the total cups of water and compare it to the goal to see if the target was achieved at the end of the day.*

Planned Inputs and Outputs:

INPUTS:

- *Name*
- *Weight in pounds*
- *Age*
- *Sex*
- *Exercise Frequency*
- *How many cups did they drink within the hour*
- *Time of the day*
- Answer to end the day (1 or 2)

OUTPUTS:

- *Welcome message with the user's name*
- *The choices for exercise frequency and sex options*
- *The questions asking how much they drank and what time it is*
- *Number of cups needed for the day*

- *Total cups drank at the end of the day*
- *Final answer saying if they drank enough water today ("Yes" or "No")*

Logic Plan (Pseudocode):

```
START

PRINT "Hello, welcome to your very own water intake tracker! (Be grateful it's free)
Please state your name."
INPUT NAME

PRINT "Hi, " + NAME + "! Before we start, please enter your details."

PRINT "Your weight in pounds:"
INPUT WEIGHT

PRINT "Your age:"
INPUT AGE

IF AGE < 14 THEN
  PRINT "This program isn't targeted towards your age range just yet!"
END IF

PRINT "Your sex (Please enter Male or Female only):"
INPUT SEX

PRINT "And how frequently you exercise (A. NEVER B. SELDOMLY C. SOMETIMES D.
FREQUENTLY E. ALWAYS):"
INPUT EXERCISE

RECOMMENDED_INTAKE = WEIGHT / 16

IF SEX = "MALE" AND AGE >= 14 AND AGE < 60 THEN
  RECOMMENDED_INTAKE += 3
ELSE IF SEX = "FEMALE" AND AGE >= 14 AND AGE < 60 THEN
  RECOMMENDED_INTAKE = RECOMMENDED_INTAKE
ELSE IF AGE >= 60 THEN
  RECOMMENDED_INTAKE = 8
END IF

IF EXERCISE = "A" THEN
  RECOMMENDED_INTAKE = RECOMMENDED_INTAKE
ELSE IF EXERCISE = "B" THEN
  RECOMMENDED_INTAKE += 1
ELSE IF EXERCISE = "C" THEN
  RECOMMENDED_INTAKE += 2
ELSE IF EXERCISE = "D" THEN
  RECOMMENDED_INTAKE += 3
ELSE IF EXERCISE = "E" THEN
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   RECOMMENDED_INTAKE += 4
END IF

PRINT "Thank you for completely answering our questions! Come back when you drink water!"

NEW_QTY = 0
CHOICE = 2  // Starts at 2 so the loop runs

WHILE CHOICE = 2 DO
   PRINT "How many cups did you drink within the hour?"
   INPUT QTY

   PRINT "What time of the day is it?"
   INPUT HOUR

   NEW_QTY += QTY

   PRINT "Do you want to end the day? 1 for Yes, 2 for No."
   INPUT CHOICE
END WHILE

PRINT "Number of cups needed: " + RECOMMENDED_INTAKE + ", Total cups drank: " + NEW_QTY + "."

IF NEW_QTY >= RECOMMENDED_INTAKE THEN
   PRINT "Did you drink enough? Yes!"
ELSE
   PRINT "Did you drink enough? No! Try hydrating yourself!"
END IF

END
```